

History and Physical Write-Up

Patient: Hannah Munguia-Flores **MRN:** 10161495501 (MGB) / 54079173 (BWH) / 10636426 (BWFH) **DOB:** 8/18/1995 | **Age:** 30 | **Legal Sex:** Female **Date of Write-Up:** August 4, 2026 **Prepared by:** Cleo (Spectator Health Clinical AI)

Introductory Statement / Chief Complaint

Ms. Hannah Munguia-Flores is a 30-year-old female with a complex multisystem illness characterized by severe, debilitating fatigue, orthostatic intolerance, post-exertional malaise (PEM), chronic urticaria, and ADHD, who is presenting for comprehensive clinical review in the context of Long COVID syndrome. Symptoms have persisted for approximately four years following initial SARS-CoV-2 infection in July 2022, significantly impairing her functional capacity and necessitating a leave of absence from her doctoral program at MIT.

Source and Reliability

History obtained from the patient (reliable historian), supplemented by Mass General Brigham (MGB) electronic medical record including visit notes from Brigham and Women's Hospital (BWH), Brigham and Women's Faulkner Hospital (BWFH), and Massachusetts General Hospital (MGH) spanning through July 2026. Patient is a trained aerospace engineering PhD student — detailed, precise, and highly engaged in her own care. No interpreter required.

Primary Care Provider: Kristen Elaine Medley, MD, MPA — Brigham Primary Physicians at Faulkner, 1153 Centre Street, Suite 5J, Jamaica Plain, MA 02130 | (617) 983-7300

History of Present Illness

Ms. Munguia-Flores was in her usual state of health through early 2022, functioning as an active aerospace engineering PhD student at MIT. In July 2022, she contracted SARS-CoV-2 for the first time. Approximately one month later, she experienced a severe mononucleosis-like illness — subsequently confirmed as EBV primary infection (EBV VCA IgM reactive in June 2025) — from which she did not recover to baseline. She developed a progressive syndrome of severe fatigue, cognitive dysfunction, and orthostatic intolerance that has continued, now nearly four years later. She has had two subsequent SARS-CoV-2 infections (August 2023, September 2024), each with approximately one week of acute symptoms and no clear lasting change to her baseline Long COVID trajectory.

Fatigue and Post-Exertional Malaise: Fatigue is her dominant, most disabling complaint. It is severe, rated 10/10 in intensity, and frequently requires her to remain in a horizontal position for extended periods. She reports that any physical or cognitive exertion beyond a low threshold reliably produces a delayed worsening of symptoms (PEM), typically manifesting 24–72 hours after the triggering activity — consistent with the pattern characterized in ME/CFS research. She tracks her recovery data via WHOOP biometric device; analysis of

774+ days of data demonstrates a consistent 3-day lag effect between activity exposure and measurable symptomatic deterioration, with WHOOP recovery score being a strong predictor. She has adhered to a structured pacing protocol and is currently under the care of a neuropsychiatrist specializing in Long COVID and ME/CFS (outside MGB).

Cognitive Dysfunction (Brain Fog): She describes a severe, qualitatively distinct cognitive impairment she rates 10/10 — characterized by memory difficulty, poor concentration, and slowed processing. She explicitly distinguishes this from her pre-existing ADHD: “This brain fog is different.” Prior to illness she was functioning at a high academic level in an elite engineering doctoral program. She has been on medical leave from MIT; one semester remains to complete her degree, and she faces a hard deadline to return in September 2026 or forfeit progress toward her degree.

Orthostatic Intolerance / Dysautonomia: She experiences racing heart rate, dizziness, and near-syncope upon transitions from supine to upright. These symptoms partially meet criteria for postural orthostatic tachycardia syndrome (POTS), though formal autonomic reflex testing (ART) performed at MGH on 2/3/2025 by Dr. Nathaniel Robbins was essentially normal — POTS criteria not met on tilt table (heart rate augmentation 15–20 bpm, below the 30 bpm threshold for adults). CASS score was 2 (mild). Patchy forearm QSART was noted, most likely attributable to medication interference (fluoxetine, naltrexone) rather than true small fiber sudomotor neuropathy. Symptoms have been partially managed with physical therapy (MGH POTS PT program, completed), high-salt diet (~3g sodium/day), salt tablets (significant symptom relief per patient report), and recently propranolol 20 mg TID (started 6/22/2026 by her neuropsychiatrist — though her PCP, Dr. Medley, flagged concern for worsening underlying dysautonomia given BP of 98/60 at her 6/25/2026 visit and recommends tapering).

Stellate Ganglion Blocks (SGBs): She underwent a series of SGBs for dysautonomia management at BWH Pain Management Clinic. Left SGB performed 5/29/2026 (Dr. Alexander Kim): patient noted transient HRV increase from 25–35 to 49, with associated left Horner’s syndrome (ptosis, miosis, facial flushing) and hoarse voice, all resolving. Right SGB performed 6/12/2026 (Dr. Kim): right eye droop and facial flushing noted, with no change in HRV. At 7/9/2026 follow-up (Dr. Petra Majdak and fellow Dr. Isaac Freedman), she reported no sustained improvement in dysautonomia from either block. She expressed interest in repeating both procedures with steroid augmentation; repeat bilateral SGBs are scheduled for August 21, 2026 and September 4, 2026 with Dr. Kim using dexamethasone.

Chronic Urticaria: She has been diagnosed with chronic urticaria, managed with omalizumab (Xolair). Monthly injections at 150 mg were not adequately controlling her hives; as of 7/17/2026, the frequency was escalated to every 14 days. She carries an EpiPen (epinephrine 0.3 mg auto-injector). She also takes cetirizine 10 mg and famotidine 20 mg as part of her urticaria regimen. Tryptase 3.0 (obtained previously) rules out systemic mastocytosis. CUI score was 6.3 (not consistent with autoimmune urticaria).

Headaches: She describes headaches occurring approximately 4 times per week, severe at 10/10, lasting approximately 6 hours. Previously described as tension-type (over the eyes), they have become more global — frontal and occipital, bilateral, with photophobia and phonophobia. No nausea, no visual aura. Acetaminophen and naproxen provide relief.

Musculoskeletal Pain: She reports neck and upper shoulder pain with a classic “coat hanger” distribution (bilateral trapezii, cervical paraspinals, extending up the lower skull), as well as bilateral SCM

pain, and bilateral hip and knee discomfort. Low-dose naltrexone (LDN 4.5 mg) was initiated approximately 5.5 months prior to her July 2026 visit and she reports it has been helpful for joint pain management.

Sleep: Disrupted sleep with approximately 3 nighttime awakenings. No difficulty initiating sleep. Daytime napping while on medical leave.

Vertigo: Intermittent faintness and vertigo with abrupt head turning, onset approximately 2 weeks prior to her 7/9/2026 visit — coinciding with bilateral ear cleaning procedures. Bilateral cerumen impaction was noted at her 6/25/2026 annual physical; ear irrigation and Debrox drops were prescribed, with ENT referral if unsuccessful.

Reproductive/Gynecologic History: Loryna (ethinyl estradiol 0.02 mg/drospirenone 3 mg) restarted 6/25/2026 for contraception; previously managed by prior PCP. Last Pap smear approximately 1 year prior (no abnormal history). No history of STIs. Reports dyspareunia. G0P0.

EBV Status: EBV VCA IgM was reactive in June 2025 (suggesting recent or reactivated infection). Repeat EBV VCA IgM on 6/25/2026 returned negative, suggesting resolution of acute reactivation phase. Dr. Medley reviewed patient's inquiry about the Yale valacyclovir protocol and found limited supporting evidence; patient is sending the Iwasaki Lab paper. HSV-1/2 IgG negative (January 2025). Monospot negative (May 2025).

Asthma: Diagnosed at the Long COVID clinic. Well-controlled for over 5 years, no current symptoms.

Iron Deficiency: History of iron deficiency anemia (ferritin 5–11 in 2024). Received IV iron infusion December 2025. Ferritin normalized to 191 ug/L on 6/25/2026 (above reference range, likely post-infusion repletion). Iron 92, TIBC 347, transferrin saturation 27% — all within normal limits.

Mental Health: She has a known history of major depressive disorder (MDD), stable for 3 years on fluoxetine 20 mg daily. She denies current depression or anxiety. Significant psychosocial stress is present related to her academic timeline and housing/financial instability (lives with boyfriend's family in Jamaica Plain, food insecurity, difficulty affording medications).

ADHD: Previously diagnosed; managed with dextroamphetamine-amphetamine (Adderall XR) 15 mg daily. She describes ADHD-related attentional difficulties as distinct from her Long COVID brain fog.

Past Medical History

1. **Long COVID Syndrome** (U09.9) — onset July 2022 following initial SARS-CoV-2 infection. Managed at MGB Long COVID Clinic, with multidisciplinary team including neuropsychiatry (outside MGB), allergy/immunology (Ramirez), and pain management (Kim, Majdak).
2. **Post-Infectious Fatigue / ME/CFS-like phenotype** — evolving from Long COVID; tracked with WHOOP biometrics over 2,292+ days.
3. **Dysautonomia / Orthostatic Intolerance** (G90.1) — clinical POTS-like presentation; formal ART essentially normal (2/3/2025, Dr. Robbins, MGH).
4. **Chronic Urticaria** — diagnosed at Long COVID Clinic; managed with Xolair (escalated to every 14 days 7/17/2026), cetirizine, famotidine. Tryptase 3.0 (mastocytosis excluded). CUI 6.3

- (autoimmune type excluded).
5. **Asthma** — well-controlled, no symptoms >5 years.
 6. **Major Depressive Disorder (MDD)** — stable on fluoxetine 20 mg daily for ~3 years.
 7. **ADHD** — managed with Adderall XR 15 mg daily.
 8. **Iron Deficiency Anemia** (D64.9) — history of ferritin 5–11 (2024); IV iron infusion 12/2025; ferritin now 191 (6/2026).
 9. **EBV Reactivation** — VCA IgM reactive June 2025; negative June 2026.
 10. **Vitamin D Deficiency** (E55.9) — 25-OH Vitamin D 49 ng/mL (6/25/2026, within normal range, likely supplemented).
 11. **Bilateral Cerumen Impaction** — treated 6/2026; ear cleaning associated with transient vertigo.
 12. **Hypothalamic-Pituitary dysregulation** — suspected, consistent with EBV-associated HPA axis disruption and LH/FSH suppression (possibly compounded by OCPs).

Past Surgical History: No surgical history documented.

Hospitalizations: No inpatient admissions documented in MGB records.

Medications

(As of July 2026, compiled from MGB records and patient-reported medications)

Medication	Dose	Route	Frequency	Indication
Fluoxetine (Prozac)	20 mg	Oral	Daily	MDD
Dextroamphetamine-amphetamine (Adderall XR)	15 mg	Oral	Daily	ADHD
Naltrexone (LDN)	4.5 mg	Oral	Daily	Pain management, LC
Omalizumab (Xolair)	150 mg/mL	Subcutaneous	Every 14 days (escalated 7/17/2026)	Chronic urticaria
Propranolol (Inderal)	20 mg	Oral	Three times daily	Orthostatic tachycardia, POTS
Ethinyl estradiol 0.02 mg / drospirenone 3 mg (Loryna)	1 tablet	Oral	Daily	Contraception
Cetirizine (Zyrtec)	10 mg	Oral	Daily	Urticaria
Famotidine (Pepcid)	20 mg	Oral	Daily	Urticaria, histamine blockade
Epinephrine (EpiPen)	0.3 mg/0.3 mL	IM auto-injector	PRN	Anaphylaxis
Tretinoin (Retin-A)	0.05% cream	Topical	As directed	Dermatologic conditions
Methylene blue	10 mg	Oral	Daily	Patient-reported mitochondrial support
Acetaminophen (Tylenol)	As needed	Oral	PRN	Headache, pain

Naproxen (Aleve)	As needed	Oral	PRN	Headache
Salt tablets	As directed	Oral	Daily	Orthostatic intolerance
CoQ10 / Alpha-lipoic acid	Various	Oral	Daily	Supplement (minimal reported benefit)

Allergies / Adverse Reactions

No Known Allergies (confirmed 6/25/2026, reviewed by Charmaine V. Griffith, MA)

Family History

- Paternal aunts and paternal grandmother: breast cancer diagnosed at ages 50 and 70; associated BRCA gene mutation (not BRCA1 or BRCA2 — specific variant unknown).
- Paternal family history: skin cancer.
- Hannah's personal BRCA status: **not yet tested**.
- No known family history of cardiac disease, diabetes, or autoimmune conditions documented in MGB records.

Social History

- **Living situation:** Lives in Jamaica Plain, MA with boyfriend's family.
- **Occupation:** PhD candidate in aerospace engineering, MIT. On medical leave; must return September 2026 or forfeit degree progress. One semester remains.
- **Education:** Graduate-level (MIT aerospace engineering PhD program). Undergraduate degree presumed; MIT grad student implies highly educated.
- **Hobbies:** Sewing, hand quilting.
- **Diet:** Mixed plant-based and omnivorous. High sodium intake (~3 g/day, per POTS protocol). Daily bagel with cream cheese; occasional red meat.
- **Social support:** Boyfriend, boyfriend's family. Some support from family.
- **Social stressors:** Food insecurity, housing instability, difficulty affording medications, significant academic pressure.
- **Tobacco:** Never smoker, no passive exposure.
- **Alcohol:** None.
- **Recreational drugs:** Not documented.
- **Sexual history:** Sexually active (heterosexual, boyfriend). Reports dyspareunia. No known STI history.
- **Advance directives:** Not documented.
- **Functional status:** Significantly impaired. Frequently bed- or couch-bound. Unable to sustain standing for prolonged periods. Limited ADLs on symptomatic days. Uses WHOOP and Visible app to track functional capacity.

Review of Systems

Constitutional: Severe fatigue (10/10), unexplained weight changes. Denies fever or chills. Requires extended horizontal rest daily.

HEENT: Bilateral ear discomfort and pain (cerumen impaction treated 6/2026). New intermittent vertigo with head turning. No epistaxis. No dental problems. No changes in smell or taste.

Eyes: No discharge, no diplopia. No vision changes.

Cardiovascular: Orthostatic racing heart rate. No chest pain. No palpitations at rest. No syncope. Managed POTS-like symptoms with salt and propranolol.

Pulmonary: Asthma — well-controlled, no current cough, dyspnea, or wheezing. Denies shortness of breath at rest.

Gastrointestinal: Reports intermittent diarrhea. Denies abdominal pain, constipation, blood in stool, or nausea.

Genitourinary: Reports dyspareunia. Denies urinary symptoms or hematuria.

Neurological: Severe brain fog (10/10), headaches 4x/week (bilateral, 10/10, photophobic/phonophobic, 6 hours duration, responds to OTC analgesics). Intermittent dizziness upon standing. Intermittent vertigo with head turning (recent). No seizures, no syncope, no paresthesias documented.

Psychiatric: MDD — stable ×3 years on fluoxetine. Denies current depression or anxiety. Significant psychosocial stressors (academic, financial, housing). ADHD managed.

Musculoskeletal: Bilateral neck/shoulder “coat hanger” pain, bilateral SCM, bilateral hip and knee discomfort. Improved with LDN.

Skin: Chronic urticaria — not fully controlled on monthly Xolair (escalated). No new rashes.

Heme/Lymph: History of iron deficiency anemia (post-infusion ferritin now elevated). No lymphadenopathy documented. No bleeding or clotting history.

Endocrine: No thyroid symptoms. TSH 3.32 (normal, 6/25/2026). No diabetes (HbA1c 5.2%). Loryna (OCP) restarted 6/25/2026.

Rheumatologic: No known connective tissue disease. Urticaria workup negative for autoimmune etiology (CUI 6.3).

Physical Examination

(Most recent complete exam: 6/25/2026, Dr. Kristen Medley, BWFH; supplemented by 7/9/2026 BWH Pain Management exam, Dr. Majdak)

Vital Signs (6/25/2026): - Blood Pressure: 98/60 mmHg (*low — flagged by Dr. Medley as likely worsened by propranolol*) - Heart Rate: 86 bpm - Temperature: 36.5°C (97.7°F), oral - Respiratory Rate: 16 breaths/min - SpO2: 99% (room air) - Height: 159.3 cm (5'2.7") - Weight: 56.2 kg (123 lb 12.8 oz) - BMI: 22.13 kg/m²

General: Well-developed, well-nourished young woman in no acute distress. Alert and cooperative. Appears well but not robustly energetic given clinical history.

HEENT: - Head: Normocephalic, atraumatic. - Eyes: No discharge bilaterally. Conjunctivae normal. Pupils equal, round, and reactive to light bilaterally. No scleral icterus. - Ears: Tympanic membranes, ear canals, and external ears normal bilaterally (6/25/2026). (*Note: bilateral cerumen impaction diagnosed and treated; subsequent 7/9/2026 note reports intermittent vertigo with head turning onset ~2 weeks after ear cleaning.*) - Nose: No discharge or obstruction noted. - Mouth/Throat: Moist mucous membranes. Oropharynx clear, no exudate or erythema.

Neck: Supple. No thyromegaly. No tracheal deviation. No cervical adenopathy. No carotid bruits. (7/9/2026: *Tenderness to palpation of bilateral trapezius and cervical paraspinal musculature.*)

Lymph Nodes: No cervical, axillary, or pectoral adenopathy.

Cardiovascular: Normal rate and regular rhythm. Carotid pulses 2+ bilaterally. Dorsalis pedis pulses 2+ bilaterally. No lower extremity edema. (*No murmurs, rubs, or gallops documented.*)

Pulmonary: Normal pulmonary effort. No tachypnea. No wheezing, rhonchi, or rales. Chest expansion symmetric.

Breasts: No swelling, mass, skin changes, discharge, or tenderness bilaterally.

Abdomen: Soft, non-distended. Bowel sounds normal. No tenderness to palpation. No masses. No hepatosplenomegaly documented.

Musculoskeletal: Cervical range of motion normal, neck supple. No lower extremity edema. (7/9/2026: *Tenderness to palpation bilateral trapezii and cervical paraspinals. Good tone throughout. No joint swelling or synovitis.*)

Skin: Warm. No acute rash at time of exam. (*Chronic urticaria present intermittently — managed with Xolair/cetirizine/famotidine.*)

Neurological: Alert and oriented to person, place, and time. No focal neurologic deficit. Motor strength — no abnormalities or asymmetry noted. Sensory intact to light touch throughout. Standard gait normal. Behavior cooperative. (7/9/2026: *CNs grossly intact at exam. No motor blockade post-SGB.*)

Psychiatric: Behavior normal. Thought content normal. Cooperative. No suicidal ideation documented.

Laboratory Results

(6/25/2026 — Dr. Medley, BWFH)

CBC: | Component | Value | Reference | Flag | |—|—|—|—| | WBC | 8.43 K/uL | 4.00-11.00 | — | | RBC | 4.22 M/uL | 4.00-5.20 | — | | Hemoglobin | 12.8 g/dL | 12.0-16.0 | — | | Hematocrit | 38.0% | 36.0-46.0 | — | | MCV | 90.0 fL | 80.0-100.0 | — | | Platelets | 324 K/uL | 150-450 | — |

Basic Metabolic Panel: | Component | Value | Reference | Flag | |—|—|—|—| | Sodium | 136 mmol/L | 136-145 | — | | Potassium | 4.1 mmol/L | 3.4-5.1 | — | | Chloride | 101 mmol/L | 98-107 | — | | CO2 | 27 mmol/L | 20-31 | — | | BUN | 10 mg/dL | 6-23 | — | | Creatinine | 0.73 mg/dL | 0.50-1.00 | — | | Glucose | 84 mg/dL | 70-99 | — | | Calcium | 9.4 mg/dL | 8.5-10.5 | — | | eGFR | 113 mL/min/1.73m² | >59 | — |

Hepatic Panel: | Component | Value | Reference | Flag | |—|—|—|—| |
AST | 20 U/L | 10-50 | — | | ALT | 10 U/L | 10-50 | — | | Alkaline
Phosphatase | 40 U/L | 40-130 | — | | Total Bilirubin | 0.2 mg/dL | 0.0-
1.2 | — | | Total Protein | 7.6 g/dL | 6.4-8.3 | — | | Albumin | 3.9 g/dL |
3.5-5.2 | — |

Lipid Panel: | Component | Value | Reference | Flag | |—|—|—|—| |
Total Cholesterol | 180 mg/dL | <200 | — | | HDL | 59 mg/dL | ≥40 | —
| | Calculated LDL | 66 mg/dL | <130 | — | | Triglycerides | **355 mg/dL**
| ≤150 | **H** | | Non-HDL Cholesterol | 121 mg/dL | — | — | | Cardiac
Risk Ratio | 3.1 | 0.0-5.0 | — |

Note: Hypertriglyceridemia (355) — notably elevated. May be related to OCPs, POTS/dysautonomia physiology, or dietary factors. Warrants follow-up.

Other: | Test | Value | Reference | Flag | |—|—|—|—| | HbA1c | 5.2% |
4.3-5.6 | — | | TSH | 3.32 uIU/mL | 0.40-5.00 | — | | 25-OH Vitamin D |
49 ng/mL | 20-50 | — | | Ferritin | **191 ug/L** | 30-150 | **H** | | Iron | 92
ug/dL | 28-170 | — | | TIBC | 347 ug/dL | 220-460 | — | | Transferrin
Saturation | 27% | 14-50 | — | | EBV VCA IgM | Negative | Negative |
— |

Procedures / Studies

Autonomic Reflex Testing (2/3/2025, MGH — Dr. Nathaniel Robbins, Director MGB Small Fiber and Autonomic Neurology Program): - QSART: Normal at 3 sites; patchy forearm low sweat volume — likely medication interference (fluoxetine, naltrexone), small fiber sudomotor neuropathy possible but less likely. - Head-Up Tilt (70°, 10 min): Baseline supine HR ~70 bpm → 85-90 bpm during HUT (augmentation ~15-20 bpm; adult threshold >30 bpm — does NOT meet POTS criteria). SBP 110 → 130 (no orthostatic hypotension). - Valsalva Maneuver: Normal phases II and IV. PRT 1.6s. Valsalva ratio 2.6 (5th %ile: 1.46) — normal. - HRDB: Normal — best HR difference 22.6 bpm (5th %ile: 14 bpm). - **CASS = 2 (mild)**. Conclusion: “Essentially normal without evidence of sudomotor dysfunction, cardiovagal or vasomotor baroreflex dysfunction, orthostatic hypotension, or excessive orthostatic tachycardia.”

Stellate Ganglion Block — Left (5/29/2026, BWH — Dr. Alexander Kim): - Indication: Dysautonomia - Medication: 9 mL 0.25% bupivacaine + 10 mg dexamethasone, ultrasound-guided to longus colli - Outcome: Positive Horner’s syndrome (ptosis, miosis, facial flushing), hoarse voice — resolved. HRV increased transiently from 25-35 → 49. No sustained benefit.

Stellate Ganglion Block — Right (6/12/2026, BWH — Dr. Alexander Kim): - Same protocol. Post-procedure: right eye droop and facial flushing, resolved. No HRV change. No sustained improvement. - Follow-up 7/9/2026: No sustained improvement from either block. Patient wishes to repeat with steroid augmentation. - **Scheduled:** Repeat bilateral SGBs — August 21 + September 4, 2026 (Dr. Kim, BWH Pain Management).

Specialist Team

Specialty	Provider	Location	Last Seen
Primary Care	Kristen E. Medley, MD, MPA	BWFH, Jamaica Plain	6/25/2026

Pain Management	Petra Majdak, MD, PhD	BWH Pain Clinic, Chestnut Hill	7/9/2026
Pain Management / Procedures	Alexander J. Kim, MD	BWH Pain Center	6/12/2026
Allergy / Immunology	Lourdes G. Ramirez, MD	BWF Allergy	7/17/2026 (Xolair)
Autonomic Neurology	Nathaniel Robbins, MD	MGH Neuromuscular	2/3/2025
Neuropsychiatry (Long COVID/ME/CFS)	Outside MGB	—	Ongoing

Summary Statement

In summary, Ms. Munguia-Flores is a 30-year-old previously high-functioning MIT aerospace engineering PhD student with a history of asthma, ADHD, and MDD who developed a debilitating multisystem post-infectious illness following SARS-CoV-2 infection in July 2022 and concurrent/subsequent EBV infection, characterized by severe PEM-associated fatigue, cognitive dysfunction, orthostatic intolerance with a POTS-like clinical picture (though autonomic reflex testing was formally normal), chronic urticaria (partially controlled), bilateral coat-hanger headache/myalgia pattern, and significant psychosocial stressors. She has failed to return to baseline after nearly four years of illness, despite adherence to pacing, POTS physical therapy, salt loading, LDN, fluoxetine, and two stellate ganglion blocks, and is currently on Xolair (escalated frequency), propranolol, Adderall XR, and Loryna. She is found to have elevated triglycerides (355), low blood pressure (98/60) possibly worsened by propranolol, and normal CBC, metabolic panel, hepatic panel, TSH, and HbA1c. The leading diagnostic framework is Long COVID with ME/CFS phenotype, hypothalamic-pituitary dysregulation, and possibly persistent gut spike protein burden as the driving pathophysiology.

Problem List

(In order of clinical priority)

1. Long COVID Syndrome / ME/CFS-phenotype with severe PEM
 2. Orthostatic Intolerance / Dysautonomia (clinical POTS, formally normal ART)
 3. Chronic Urticaria — partially controlled
 4. Recurrent severe headaches (possible tension-type vs. new migrainous pattern)
 5. Cognitive Dysfunction (brain fog) — severe, functional impact
 6. Musculoskeletal pain — bilateral coat-hanger pattern, hip/knee
 7. Hypertriglyceridemia (newly identified, 355 mg/dL)
 8. Hypotension on propranolol (BP 98/60) — medication safety concern
 9. Post-iron infusion elevated ferritin (191, above reference)
 10. Major Depressive Disorder — currently stable
 11. ADHD — currently managed
 12. Asthma — well-controlled
 13. Family history of breast cancer with BRCA mutation — BRCA testing not yet done
 14. Academic/psychosocial pressure (deadline September 2026)
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Assessment and Plan

1. Long COVID Syndrome / ME/CFS-phenotype

Assessment: This patient's presentation is most consistent with Long COVID with ME/CFS phenotype, characterized by: (1) post-infectious onset following SARS-CoV-2 + EBV co-infection, (2) severe PEM with 3-day lag demonstrated biometrically, (3) multisystem involvement, (4) autonomic dysregulation, and (5) near-complete failure to recover over 4 years. The leading mechanistic hypothesis, supported by emerging literature (Freire 2026, Iwasaki Lab), is persistent viral antigen/spike protein burden in gut and other tissues, driving ongoing immune dysregulation, HPA axis disruption, and microbiome perturbation. Alternative or co-contributing diagnoses include: - **Persistent EBV reactivation:** VCA IgM was reactive June 2025, negative June 2026 — acute phase resolved; latent EBV with potential immunomodulatory effect persists. Valacyclovir protocol (Yale/Iwasaki) remains a pending discussion pending patient-submitted evidence. - **ME/CFS (independent of COVID):** The severity and biometric pattern (PEM, 3-day lag, Ghali 2023 pacing adherence data) fulfills ME/CFS criteria per ICC/CCC. GET is contraindicated. - **Small fiber neuropathy:** Less likely given normal formal ART, but patchy QSART forearm is of uncertain significance. - **Mast cell activation syndrome (MCAS):** Tryptase 3.0 makes systemic mastocytosis unlikely; MCAS (without elevated tryptase) remains possible and may account for urticaria + multisystem symptoms. DO NOT MISS.

Plan: - Continue pacing protocol (WHOOP-guided Repair Spectrum Framework). Current prescription: strain <2, PacePoints <6. - Continue LDN 4.5 mg (helping pain). - Discuss fluvoxamine per REVIVE-TOGETHER trial (Reis 2026, AIM) — FSS reduction -0.58 at day 90; benefits wane off-drug, likely requires maintenance dosing. Present to treating neuropsychiatrist. - Follow up on valacyclovir paper — if Dr. Medley remains opposed, consider referring to ID/Virology for second opinion given EBV history. - Consider referral for formal ME/CFS evaluation if not already complete. - Patient education: reinforce that GET is contraindicated; heart rate monitoring at/below anaerobic threshold is the tool (PACELOC 2025 — 15% weekly PEM reduction with structured pacing). - Follow up in 1-2 months; repeat WHOOP data review.

2. Orthostatic Intolerance / Dysautonomia

Assessment: Clinical picture consistent with dysautonomia despite normal formal ART (2/3/2025). The disconnect between clinical symptoms and formal testing may reflect: (1) medication effect on ART (fluoxetine, naltrexone noted to affect QSART), (2) intermittent/fluctuating autonomic dysfunction not captured at time of formal testing, or (3) central autonomic dysregulation not detected by peripheral testing. Current BP 98/60 with propranolol 20 mg TID is a patient safety concern — this medication may be worsening hypotension and her overall dysautonomia. SGBs provided transient HRV benefit but no sustained improvement.

Plan: - **Urgent:** Communicate with prescribing neuropsychiatrist to taper propranolol. Dr. Medley's recommendation to taper is appropriate given BP 98/60. Reassess BP and orthostatic symptoms off propranolol. - Continue salt tablets and high-sodium diet (~3 g/day) — patient reports significant symptom relief. - Repeat bilateral SGBs scheduled: August 21 and September 4, 2026 (Dr. Kim, BWH). Proceed as planned with steroid augmentation. - Continue monitoring HRV via WHOOP — useful surrogate for autonomic state. - If propranolol tapered and HR remains problematic, consider low-dose

ivabradine (if-channel blocker) as a POTS alternative with less hypotensive effect. Discuss with cardiology or neuropsychiatrist. - Consider formal cardiology evaluation (tilt table repeat off medications) if symptoms worsen.

3. Chronic Urticaria

Assessment: Chronic urticaria not adequately controlled on monthly Xolair 150 mg. Escalation to every 14 days appropriate (initiated 7/17/2026). CUI 6.3 and tryptase 3.0 argue against autoimmune and mast cell systemic disease. MCAS remains possible. Current regimen: Xolair + cetirizine + famotidine. EpiPen appropriate given risk of angioedema/anaphylaxis.

Plan: - Continue Xolair 150 mg every 14 days (Dr. Ramirez managing). - Continue cetirizine 10 mg daily and famotidine 20 mg daily. - Ensure EpiPen available at all times; reinforce anaphylaxis action plan. - Reassess CIU activity with formal UAS7 scoring at next allergy visit. - If urticaria uncontrolled at q14d Xolair, discuss 300 mg dosing or addition of cyclosporine per allergy guidelines. - Consider MCAS workup (24h urine prostaglandins, plasma histamine, urinary methylhistamine) if suspicion increases. - Follow up with Dr. Ramirez (next: 7/17/2026; future scheduled).

4. Headaches

Assessment: Headaches 4x/week, bilateral, severe, photophobic/phonophobic, 6-hour duration, responding to acetaminophen and naproxen. The evolution from tension-type to more global with light/sound sensitivity raises question of migraine without aura (by ICHD-3 criteria: ≥ 5 attacks, 4-72h, bilateral or unilateral, with photophobia/phonophobia). The coat-hanger pattern and cervical/trapezius tenderness suggest myofascial/tension component as well. Important: frequent NSAID/analgesic use (4x/week) risks medication overuse headache (MOH). DO NOT MISS.

Plan: - Obtain headache diary (frequency, duration, triggers, medications used). - Assess for MOH — if analgesics used >10 -15 days/month, counseling on limit and potential withdrawal required. - If migraine criteria met, initiate preventive therapy — topiramate or amitriptyline are first line; note amitriptyline would have dual benefit for sleep and pain (coordinate with neuropsychiatrist given existing fluoxetine — avoid serotonin syndrome risk). - Refer to neurology for formal headache evaluation if pattern worsens or migraine prevention needed. - Physical therapy for cervical/trapezius myofascial component (coat hanger pain contributing).

5. Hypertriglyceridemia

Assessment: Triglycerides 355 mg/dL — significantly elevated (normal <150). At this level, risk of pancreatitis is emerging (typically >500 mg/dL for high risk, but warrants attention). Possible contributors: Loryna (OCPs are a known cause of hypertriglyceridemia, particularly drospirenone-EE combinations), POTS physiology, high-sodium diet (may not be contributory), or other metabolic factors. LDL 66 and HDL 59 are favorable; overall cardiac risk 3.1. HbA1c 5.2% — no diabetes.

Plan: - Discuss with Dr. Medley: consider monitoring fasting triglycerides off OCPs (if clinically appropriate) to assess Loryna contribution. - Dietary counseling: reduce refined carbohydrates, sugar, and alcohol (currently none reported). - Recheck fasting lipids

in 3 months. - If triglycerides remain >500 or worsen, consider referral to endocrinology or lipid specialist; omega-3 fatty acid therapy (icosapentaenoic acid) may be considered.

6. Family History / BRCA

Assessment: Paternal aunts and grandmother with breast cancer at ages 50 and 70, with an unspecified BRCA mutation (not BRCA1 or BRCA2). Hannah's own BRCA status unknown. By NCCAP/USPSTF guidelines, a formal genetic counseling referral is indicated given this family history pattern.

Plan: - Refer for genetic counseling (BRCA and broader hereditary breast/ovarian cancer panel assessment). - Determine age of earliest relative's breast cancer diagnosis to set mammogram start date (typically 10 years before earliest diagnosis). - Counsel patient on implications of any positive result.

7. MDD / ADHD (currently stable)

Assessment: MDD stable on fluoxetine 20 mg for 3 years. ADHD managed with Adderall XR 15 mg. Significant psychosocial stressors present but patient denies active depression or anxiety.

Plan: - Continue fluoxetine 20 mg daily. - Continue Adderall XR 15 mg daily (confirm script current with neuropsychiatrist). - Monitor mood closely given acute stressors (MIT deadline, housing, finances). - Ensure neuropsychiatrist is aware of propranolol tapering plan. - Screen for suicidality at each visit given stressors.

8. Asthma (well-controlled)

Assessment: No current symptoms for >5 years. Well-controlled.

Plan: - No rescue or controller medication currently listed. Confirm patient has albuterol inhaler PRN. Add to medication reconciliation. - Continue monitoring; re-evaluate if symptoms recur.

References

1. Reis G, et al. "Fluvoxamine for Long COVID Fatigue: REVIVE-TOGETHER Randomized Trial." *Ann Intern Med*. 2026. (REVIVE-TOGETHER)
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4. Robbins N. "Autonomic Reflex Testing Report — Hannah Munguia-Flores." MGH Neurodiagnostic Autonomic Laboratory. 2/3/2025.

This write-up was prepared from MGB medical records and existing clinical knowledge. It is intended as a clinical documentation tool — not a substitute for clinician evaluation. All treatment decisions should be made by the treating team in conjunction with the patient.

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